

I MADE ARYA PRADITHA

COMPUTER SYSTEMS STUDENT | IoT | EMBEDDED SYSTEMS | MACHINE LEARNING | CREATIVE TECHNOLOGY

Kintamani, Bangli, Bali, Indonesia • 085931152688 • aryaditha13@gmail.com

PROFESSIONAL PROFILE

Active Computer Systems student at ITB STIKOM Bali with hands-on project experience in Internet of Things (IoT), embedded systems, ESP32 and Arduino development, machine learning, and creative technology. Experienced in building and testing hardware prototypes, integrating sensors and microcontrollers, developing web-based control systems, and working with image-based machine learning models. Also interested in videography and visual documentation for technology projects.

EDUCATION

Bachelor's Degree in Computer Systems (Active Student)

ITB STIKOM Bali

Relevant areas of study and practice include embedded systems, Internet of Things, microcontrollers, programming, machine learning, and technology-based system development.

PROJECT EXPERIENCE

Kosabangsa – Smart Farming & Village Cooperative System

Student Project Team Member | IoT and Web Development

- Assisted in developing an IoT-based smart farming solution using Raspberry Pi and soil moisture sensors.
- Supported soil condition monitoring and automatic irrigation system development.
- Assisted with backend and website development for the project ecosystem.
- Supported IoT integration with the Kinta Garden application for monitoring and control.
- Participated in documentation, field implementation, and system testing activities.

Smart Trash Bin

IoT Prototype and System Development

- Assisted in designing the concept and system for an IoT-based smart waste management prototype.
- Supported prototype creation, device integration, testing, and feature development.
- Participated in documenting the project's development process and progress.
- Worked collaboratively during the prototype and system development phase.

Embedded Machine Learning – Trash Classification

Machine Learning and Image Classification Project

- Assisted in preparing and organizing waste image datasets.
- Performed image preprocessing before model training.
- Assisted in building and training a Convolutional Neural Network (CNN).
- Participated in testing, accuracy analysis, performance evaluation, and model development.

ESP32-Based Car Controller

Embedded Systems and Web-Based Control Project

- Assembled a four-wheel robotic car prototype with a car frame, wheels, and DC gearmotors.

- Used ESP32 as the central microcontroller and integrated it with motor drivers.
- Developed a web-based control interface for forward, backward, left, right, and stop movement.
- Performed hardware wiring, power integration, ESP32 programming, testing, and troubleshooting.

Creative Videography & Technology Documentation

Project Documentation and Visual Content

- Documented technology project activities and IoT equipment installation processes.
- Created visual content combining technical documentation with creative storytelling.
- Developed practical skills in camera operation, composition, and video editing.

TECHNICAL SKILLS

IoT & Embedded Systems

ESP32, Arduino, Raspberry Pi, sensors, microcontrollers, hardware prototyping, wiring, testing and troubleshooting

Programming & Development

Python, Arduino IDE, basic web-based control development, system integration

Machine Learning

Image datasets, preprocessing, TensorFlow, Keras, CNN, model training and evaluation

Creative Technology

Videography, project documentation, visual storytelling, video editing

ADDITIONAL STRENGTHS

Team collaboration • Hardware assembly • System testing • Troubleshooting • Technical documentation • Communication • Creative problem solving • Adaptability

Portfolio: Project experience and technical work are available through the personal digital portfolio.